POUYA AZARANDAZ

Materials & Additive Manufacturing Researcher | Porous Structures, Characterisation & Simulation

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PROFESSIONAL SUMMARY

Mechanical and materials engineer whose M.Sc. research centred on additive manufacturing of porous, structured architectures and their experimental characterisation. Completed an M.Sc. thesis at Politecnico di Torino combining laser powder bed fusion of functionally graded porous structures, X-ray computed tomography of porosity and defects, standardised compression testing, and finite element modelling, published as a peer-reviewed journal article and presented at an MDPI international conference. Additional research experience in mechanical characterisation of advanced ceramics. Strong foundation in design for additive manufacturing, porosity and structure control, structure to property correlation, and processing science. Seeking a doctoral position applying porous-material and additive-manufacturing methods to structured sorbents for environmental and gas-cleaning applications.

CORE TECHNICAL SKILLS

Additive Manufacturing & DfAM: Laser powder bed fusion (L-PBF), design for additive manufacturing, structured and lattice (TPMS) geometry design, functionally graded porosity, build orientation and support strategy, nTopology

Porous Materials & Structure–Property: Porosity and defect quantification, relative density, designed void fraction, Gibson–Ashby scaling, functionally graded architectures, structure to property to performance correlation

Characterisation & Testing: X-ray computed tomography (XCT/CT), SEM fractography, quasi-static compression (ISO 13314), 3-point bending, Vickers indentation, fracture toughness, specimen preparation

Simulation & Modelling: ABAQUS (Explicit and Standard) FEA, Ansys Mechanical, nonlinear and contact modelling, mesh strategy, simulation-to-experiment validation

Materials: AlSi10Mg, advanced ceramics, Ti-6Al-4V, steel; metallic and ceramic materials, porosity and defect analysis, microstructural evaluation

Programming & Data: Python, MATLAB, KNIME; supervised machine learning for process and parameter optimisation, engineering data analysis

Documentation & Communication: Peer-reviewed manuscripts, technical and test reports, ISO/ASTM-referenced documentation, journal peer review; English C1

SELECTED RESEARCH & PROJECTS

Additive Manufacturing & Characterisation of Porous TPMS Structures — *M.Sc. Thesis, Politecnico di Torino* 2023 – 2025

- Designed uniform and functionally graded porous TPMS architectures (FRD, Gyroid, Fischer-Koch-S) and fabricated them by laser powder bed fusion from AlSi10Mg, applying design-for-AM constraints on feature size, overhangs, and build orientation.
- Quantified internal porosity and manufacturing-induced defects by X-ray computed tomography, separating intentionally designed void fraction from defect volume ratio across five topologies.
- Evaluated structure to property relationships through ISO 13314 quasi-static compression, Gibson–Ashby scaling, and SEM fractography, demonstrating that topology and porosity grading, not relative density alone, govern stiffness, strength, and energy absorption.
- Built an Abaqus/Explicit finite element model of a graded structure that reproduced the experimental force–displacement response ($R^2 = 0.853$), correlating simulation with measured collapse behaviour.
- Published the work as a peer-reviewed journal article and presented it at the 4th International Online Conference on Materials (MDPI); further manuscripts in preparation.

Machine Learning for Additive Manufacturing Parameter Optimisation — *Academic Project* 2024 – 2025

- Applied supervised machine-learning algorithms (Python, MATLAB, KNIME) to L-PBF powder bed fusion datasets to support process-parameter optimisation; manuscript in preparation.

Scoping Review: Porous & Lattice Structures for Implants — *Research Collaboration* 2025 – 2026

- Contributing to a systematic review of porous and lattice structures, covering design for additive manufacturing, porosity, osseointegration, and material selection (Ti-6Al-4V, NiTi).

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant — Mechanical Characterisation of Ceramics *Politecnico di Milano, Italy* Jul 2024 – Oct 2024

- Characterised flexural strength and fracture toughness of advanced ceramic specimens by 3-point bending and Vickers indentation, following ASTM-referenced test protocols.
- Executed specimen preparation, measurement campaigns, and data analysis to evaluate the structural performance of novel ceramic formulations; authored test reports ensuring reproducibility and data traceability.

Teaching Assistant — Fluid Machinery *Politecnico di Torino, Italy* 2024 – 2025

- Supported 60+ undergraduate students across laboratory sessions and problem-solving tutorials in fluid machinery and engineering thermodynamics, strengthening technical communication in an international setting.

Mechanical Design Engineer *Peyman Modiriat Saze Co., Tehran, Iran* Sep 2021 – Jul 2023

- Designed SolidWorks 3D assemblies and produced GD&T-annotated 2D drawings for industrial mechanical equipment, managing CAD documentation through iterative design reviews and systematic version control.

EDUCATION

M.Sc. Mechanical Engineering *Politecnico di Torino, Italy* 2022 – 2025

Thesis: A Comparative Multi-Method Study of Uniform and Graded TPMS Lattices Fabricated via Additive Manufacturing. Focus: L-PBF additive manufacturing, porous/TPMS design, FEA (ABAQUS), compression testing, and XCT-based characterisation.

B.Sc. Mechanical Engineering *Shahid Bahonar University of Kerman, Iran* 2017 – 2021

Graduated top 5% in department; merit scholarship for direct M.Sc. admission. Thesis: Experimental and numerical crushing analysis of a novel multibody thin-walled circular tube under axial impact loading.

PUBLICATIONS & PROFESSIONAL SERVICE

Journal: Mozafari, A., Azarandaz, P., Darijani, H., Shahsavari, H. (2022). Experimental and numerical crushing analysis of a novel multibody thin-walled circular tube under axial impact loading. *International Journal of Applied Mechanics, 14(09), 2250042.*

Conference: A Comparative Multi-Method Study of Uniform and Graded TPMS Lattices Fabricated via Additive Manufacturing. 4th International Online Conference on Materials, MDPI.

In preparation: (1) Functionally graded TPMS lattice study (compression testing, XCT, FEA); (2) Supervised ML for L-PBF parameter optimisation.

Peer reviewer: Thin-Walled Structures (Elsevier) — structural mechanics and engineering materials (2 reviews, 2023–2024).

CERTIFICATIONS

Design for Additive Manufacturing — *Arizona State University (Coursera)*

Intro to Crystal Plasticity Modeling in Ansys Mechanical — *Ansys*

Supervised Machine Learning: Regression and Classification — *DeepLearning.AI & Stanford University (Coursera)*

SOLIDWORKS Assemblies & Part Modelling (Exam Level) — *Dassault Systèmes (Coursera)*

MATLAB Onramp; Deep Learning Onramp; Regression with Deep Learning — *MathWorks Training Services*

AWARDS & LANGUAGES

Top 5% among graduates, Mechanical Engineering, Shahid Bahonar University of Kerman; merit-based M.Sc. scholarship (direct admission).

Languages: Persian (native) • English C1 (professional) • Italian A2 (developing)